

Anti-diarrhoeal activity of *Butea monosperma* in experimental animals

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Abstract

The anti-diarrhoeal potential of the ethanolic extract of stem bark of *Butea monosperma* (Lam) Kuntz has been evaluated using several experimental models in Wistar albino rats. The extract inhibited castor oil induced diarrhoea and PGE₂ induced enteropooling in rats; it also reduced gastrointestinal motility after charcoal meal administration. The results obtained establish the efficacy and substantiate the use of this herbal remedy as a non-specific treatment for diarrhoea in folk medicine.

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1. Introduction

In developing countries, a quarter of infant and childhood mortality is related to diarrhoea (Jousilahti et al., 1977). Many plants conveniently available in India are used in traditional folklore medicine for the treatment of diarrhoea and dysentery (Chopra et al., 1956). The plant *Butea monosperma* (Lam) Kuntz (Fabaceae), also known as Bastard Teak, is a medium sized tree native of the mountainous regions of India and Burma (Anonymous, 1988). The stem bark of this plant is used in indigenous medicine for the treatment of dyspepsia, diarrhoea, dysentery, diabetes, ulcers, sore throat and snake bites (Jayaweera, 1981; Varier, 1993). Although many chemical constituents such as, kino-tannic acid, gallic acid and proanthocyanidins have been isolated from the stem bark and gum (Seshadri and Trikha, 1972; Nadkarni and Nadkarani, 1976), together with (–) medicarpin from the stem bark (Bandara et al., 1989), and pterocarpan, phenols and lipids from the stem (Mishra et al., 2000; Shukla et al.,

2002), we wish to report therein on the anti-diarrhoeal effect of the ethanolic extract of the stem bark of the plant.

2. Materials and methods

2.1. Plant material

Stem bark of *Butea monosperma* was collected from the Valasa Malai in the Dharmapuri district of Tamilnadu, India in November 2002. It was identified by Professor P. Jayaraman, Taxonomist, Plant Anatomy Research Centre, Chennai. A voucher specimen (GPT/19) has been deposited in the Herbarium Section of the Department of Pharmacognosy, K.P. College of Pharmacy, Thiruvannamalai, Tamilnadu, for future reference. The stem bark (1.5 kg) was air dried and pulverized using a mechanical grinder.

2.2. Preparation of extract

The powdered plant material (750 g) was subjected to maceration process with 95% ethanol at room temperature. After exhaustive extraction, the ethanolic extract (EBM) was

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concentrated under reduced pressure at 50–55 °C. A reddish-brown coloured residue was obtained (93 g; yield 12.4%, w/w) and stored in desiccator. For pharmacological studies, a weighed amount of the dried extract was suspended in a 2% (w/v) aqueous acacia solution.

2.3. Animals used

Swiss albino mice (30–40 g) and Albino (Wistar) rats (210–235 g) of either sex were maintained at uniform laboratory conditions in standard polypropylene cages and provided with food and water ad libitum. The animals were acclimatized for a period of 14 days prior to performing the experiments.

2.4. Acute toxicity study

Swiss albino mice were divided into eight groups of six individuals. The extract was administered orally at doses ranging from 0.1 to 5 g/kg following a standard method (Turner, 1965). A group of animals treated with 2% (w/v) aqueous acacia suspension (vehicle control). The animals were continuously observed for 2 h to detect changes in the autonomic or behavioural responses. Mortality in each group was observed for 7 days. The doses of 200, 400 and 800 mg/kg were selected based on the results of preliminary toxicity testing.

2.5. Castor oil induced diarrhoea in rats

The method reported by Awouters et al. (1978), with modifications, has been used in the present study. Rats of either sex (210–235 g) were fasted for 18 h; they were then divided into five groups of five individuals. The EBM extract was administered orally at doses of 200, 400 and 800 mg/kg by gavage as suspension to the first three groups of animals. The fourth group received loperamide (3 mg/kg) orally as suspension (positive control). The fifth group, which served as the blank, was administered with aqueous acacia suspension. After 60 min of treatment, the animals of each group received 1 ml of castor oil orally, by gavage, and the consistency of faecal material and the frequency of defecation was noted up to 4 h in the transparent plastic dishes placed beneath the individual rat cages (Gnansekar and Perianayagam, 2004).

2.6. Gastrointestinal motility tests

Rats were fasted for 18 h and then placed in five cages containing five individuals in each cage. Each animal was administered orally with 1 ml of charcoal meal (5% deactivated charcoal in 10% aqueous tragacanth), followed by oral administration of EBM suspension to three groups of animals in doses of 200, 400 and 800 mg/kg. The fourth group received atropine (0.1 mg/kg, i.p.), the standard drug for comparison and the fifth group was treated with aqueous acacia suspension (vehicle control). Thirty minutes later, each ani-

mal was sacrificed and the intestinal distance moved by the charcoal meal from the pylorus was cut, measured, and expressed as a percentage of the distance from the pylorus to caecum for each animal (Mukherjee et al., 1998).

2.7. PGE₂-induced enteropooling

In this method, rats were deprived of food and water for 18 h and placed in five cages, with five animals per cage. The first three groups were treated with 200, 400 and 800 mg/kg doses of EBM. The fourth group was treated with 1 ml of a 5% (v/v) ethanol in normal saline (i.p.) and then it was treated with aqueous acacia suspension, which served as vehicle control. Immediately after the extract administration PGE₂ (Astra Zeneca, India) was administered orally to each rat (100 µg/kg) in the first three groups. The fifth group was treated with PGE₂ (100 µg/kg) as well as with aqueous acacia suspension and served as the PGE₂ control group. After 30 min following administration of PGE₂, each rat was sacrificed and the whole length of the intestine from the pylorus to the caecum was dissected out, its content collected in a test tube, and the volume measured (Mukherjee et al., 1998).

2.8. Statistical analysis

All data was expressed as the mean ± S.E.M. Statistical significance testing was performed by Student's *t*-test; *P* < 0.05 imply significance.

3. Results

The presence of steroids, flavonoids, phenolic compounds, tannins and glycosides were detected on preliminary phytochemical screening of the dried extract. The EBM extract, when orally administered in the dose range 0.1–5.0 g/kg to mice did not produce any significant changes in the autonomic or behavioural response during the observation period. No mortality was observed up to 7 days of monitoring. Hence, the extract seem to be safe for administration up to 5.0 g/kg.

The EBM extract significantly inhibited the frequency of defaecation, when compared to untreated control rats (Table 1); the activity was similar to that of loperamide, the standard anti-diarrhoeal agent. The extract also reduced the wetness of faecal droppings. The EBM extract at doses of 400 and 800 mg/kg decreased the propulsion of charcoal meal through the gastrointestinal tract, as compared with the control group. Atropine (0.1 mg/kg) reduced the motility of the intestine to a greater extent (*P* < 0.001) (Table 2). The extract (EBM) significantly inhibited PGE₂ induced enteropooling in rats in higher dose levels (Table 3). PGE₂ induced a significant increase in the fluid volume of the rat intestine when compared with control animals, received ethanol in normal saline.

Table 1
Effect of stem bark extract (EBM) on castor oil induced diarrhoea in rats

Oral pre-treatment at 0 h + castor oil at 1 h	Mean defecations in 4 h	Mean number of wet faeces in 4 h
Control (acacia suspension 5 ml/kg)	3.60 ± 0.50	3.00 ± 0.70
Standard (loperamide 3 mg/kg)	0.80 ± 0.58 ^c	0.00 ^c
EBM (200 mg/kg)	2.4 ± 0.24 ^b	1.20 ± 0.80 ^a
EBM (400 mg/kg)	1.8 ± 0.42 ^c	0.80 ± 0.52 ^b
EBM (800 mg/kg)	1.00 ± 0.48 ^c	0.40 ± 0.24 ^b

Results are mean ± S.E.M., $n = 5$. Statistical significance test with control was done by Student's t -test.

^a $P < 0.05$.

^b $P < 0.01$.

^c $P < 0.001$.

Table 2
Effect of EBM extract on small intestinal transit

Oral pre-treatment	Movement of charcoal meal (%)
Control (acacia suspension 5 ml/kg)	87.88 ± 1.84
Standard (atropine 0.1 mg/kg)	29.12 ± 0.98 ^b
EBM (200 mg/kg)	86.70 ± 0.84
EBM (400 mg/kg)	75.32 ± 1.42 ^a
EBM (800 mg/kg)	30.18 ± 1.12 ^b

Results are mean ± S.E.M., $n = 5$. Statistical significance test with control was done by Student's t -test.

^a $P < 0.01$.

^b $P < 0.001$.

Table 3
Anti-enteropooling effect of EBM extract in rats

Treatment	Volume of intestinal fluid (ml)	P -values
Ethanol in saline	0.98 ± 0.30	–
PGE ₂ in ethanol (100 µg/kg)	2.82 ± 0.26	0.001 ^a
EBM (200 mg/kg)	2.78 ± 0.14	–
EBM (400 mg/kg)	2.12 ± 0.12	0.05 ^b
EBM (800 mg/kg)	1.72 ± 0.17	0.001 ^b

Results are mean ± S.E.M., $n = 5$. Statistical significance test with control was done by Student's t -test.

^a With respect to ethanol in saline treatment.

^b With respect to PGE₂ treatment.

4. Discussion

In the present study, the extract of *Butea monosperma* exhibited significant anti-diarrhoeal activity against castor oil induced diarrhoea in rats. The extract had a similar activity as loperamide, when tested at 400 and 800 mg/kg and significantly inhibited the frequency of defecation and the wetness of the faecal droppings when compared to control rats. Castor oil releases ricinoleic acid which induces changes in mucosal fluid and electrolyte transport that results in a hypersecretory response and diarrhoea (Ammon et al., 1974; Gaginella et al., 1975). The experimental studies in rats demonstrated a significant increase in the portal venous PGE₂ concentration following oral administration of castor oil (Luderer et al., 1980). Ricinoleic acid markedly increased the PGE₂ content in the gut lumen and also caused an increase of the net secretion of the water and electrolytes into the small intestine (Beubler and Juan, 1979). Inhibitors of prostaglandin biosyn-

thesis delayed castor oil induced diarrhoea (Awouters et al., 1978).

The extract appears to act on all parts of the intestine. Thus, it reduced the intestinal propulsive movement in the charcoal meal treated model; at 800 mg/kg EBM showed activity similar to that of atropine. The extract (EBM) at different dose levels 400 and 800 mg/kg significantly inhibited the PGE₂ induced intestinal fluid accumulation (enteropooling). These observations tend to suggest that the EBM extract at doses of 400 and 800 mg/kg reduced diarrhoea by inhibiting gastrointestinal motility and PGE₂ induced enteropooling.

The present results indicate that the ethanolic extract of *Butea monosperma* possesses significant anti-diarrhoeal activity due to its inhibitory effect both on gastrointestinal propulsion and fluid secretion. The inhibitory effect of the extract justifies the use of the plant as a non-specific anti-diarrhoeal agent in folk medicine.

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